Training dynamics of tree tensor networks: saddle-to-saddle, lazy and rich regimes

TTN regimes project*

Draft of October 9, 2026

Abstract

[Abstract to be written last: question, model (TTN regression with exact population loss), the regimes found and the parameters that select them, main new results with their status.]

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*Working draft; authors to be added.

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Conventions. Every claim carries a status tag: proven (proof given or cited), derived (derivation under stated assumptions), empirical (supported by experiments only) or conjecture (stated without sufficient derivation or evidence). Results we believe to be new are marked new, known results known with a reference; unmarked statements are definitions or standard background. Notation is defined in Section 2 and collected in `macros.tex`.

1 Introduction and summary of findings

[Skeleton. To be written once the results sections exist; intended content is in the TODO comments of this file.]

1.1 Question and motivation

[Which training regimes do tree tensor networks show under gradient methods, which parameters select them, and how far theory predicts them; the main focus is saddle-to-saddle dynamics.]

1.2 Summary of findings

[One table with one row per question (Q1–Q8 of the research plan): finding, status tag, new or known, and where it is shown.]

1.3 Relation to prior work

[What is known (deep linear networks, diagonal networks, tensor factorization, lazy versus rich, tensor networks for learning) and what is new here.]

1.4 Outline

[Roadmap of the report.]

2 Setup: model, data, teacher, training and diagnostics

This section fixes the model studied throughout and the measurements used to classify training regimes. Every experiment uses these definitions; deviations are stated where they occur.

2.1 Tree tensor network regression

Inputs. An input x consists of N sites. Site $i \in \{1, \dots, N\}$ carries a local feature vector $\phi_i(x) \in \mathbb{R}^{d_i}$ (by default $d_i = d$). Unless stated otherwise $\phi_i(x) = x^{(i)}$ with $x^{(i)} \sim \mathcal{N}(0, \Sigma_i)$ independent across sites, and the inputs are *whitened*, $\Sigma_i = I$. We write

$$\Phi(x) = \phi_1(x) \otimes \cdots \otimes \phi_N(x) \in \mathbb{R}^{d_1 \times \cdots \times d_N}, \qquad \Sigma = \Sigma_1 \otimes \cdots \otimes \Sigma_N.$$

Trees and cores. A tree tensor network (TTN) is specified by a rooted tree $\mathcal{T}$ with node set $\mathcal{V}$, $|\mathcal{V}| = K$, and an assignment of every site to exactly one node. For a node v, S_v is the (possibly empty) set of sites attached to v, $\mathrm{ch}(v)$ the set of its children and $p(v)$ its parent. Every non-root node v is joined to its parent by the edge (bond) e_v of *bond dimension* χ_v; for the root we set $\chi_{\mathrm{root}} = 1$. Node v carries a *core* G_v, a tensor with one *physical leg* of dimension d_i for every $i \in S_v$, one leg of dimension χ_c for every child $c \in \mathrm{ch}(v)$, and one leg of dimension χ_v for its parent bond (for

the root, the scalar output). The legs other than the parent leg are the *child legs*; their dimensions multiply to the *fan-in*

$$F_v = \prod_{i \in S_v} d_i \prod_{c \in \mathrm{ch}(v)} \chi_c,$$

so G_v can be viewed as an $F_v \times \chi_v$ matrix. The parameters are $\theta = (G_v)_{v \in \mathcal{V}}$, with $\sum_v F_v \chi_v$ entries in total.

Model. The tensor $W(\theta) \in \mathbb{R}^{d_1 \times \cdots \times d_N}$ is the contraction of all cores over all bonds, and the model is

$$f_\theta(x) = \langle W(\theta),\, \Phi(x) \rangle. \tag{1}$$

Equivalently, f_θ is computed by passing messages $m_v(x) \in \mathbb{R}^{\chi_v}$ from the leaves to the root,

$$m_v(x)_a = \sum_{(j_i),(b_c)} G_v\big[(j_i)_{i \in S_v}, (b_c)_{c \in \mathrm{ch}(v)}, a\big] \prod_{i \in S_v} \phi_i(x)_{j_i} \prod_{c \in \mathrm{ch}(v)} m_c(x)_{b_c}, \qquad f_\theta(x) = m_{\mathrm{root}}(x). \tag{2}$$

Two structural facts drive the dynamics. The model is multilinear in the site features, and W is multilinear in the cores (each core appears once), so $\theta \mapsto W(\theta)$ is a homogeneous polynomial of degree K: rescaling the cores, $G_v \mapsto c_v G_v$, rescales W by $\prod_v c_v$.

Bipartitions and bond spectra. Let A_v be the set of sites in the subtree rooted at v. The edge e_v splits the sites into A_v and its complement, and $W^{\langle e_v \rangle}$ denotes the matricization of W with the sites in A_v as row indices. Contracting the subtree below v and the rest of the network separately gives a factorization $W^{\langle e_v \rangle} = T_v C_v^\top$ with $T_v \in \mathbb{R}^{(\prod_{i \in A_v} d_i) \times \chi_v}$, so $\operatorname{rank} W^{\langle e_v \rangle} \le \chi_v$ (Hackbusch and Kühn, 2009; Grasedyck, 2010). The *bond spectrum* (Schmidt values) of an edge $e = e_v$ is the list $s_1^e \ge s_2^e \ge \cdots \ge s_{\chi_v}^e \ge 0$ of the χ_v largest singular values of $W^{\langle e \rangle}$. All nonzero singular values are among them, so $\sum_k (s_k^e)^2 = \|W\|_F^2$ on every edge. With anisotropic inputs we use function-space spectra, the singular values of $(\Sigma^{1/2} W)^{\langle e \rangle}$ with $\Sigma^{1/2} = \Sigma_1^{1/2} \otimes \cdots \otimes \Sigma_N^{1/2}$. Bond spectra are computed from the cores by QR sweeps, without forming W, to about 10^{-15} relative accuracy.

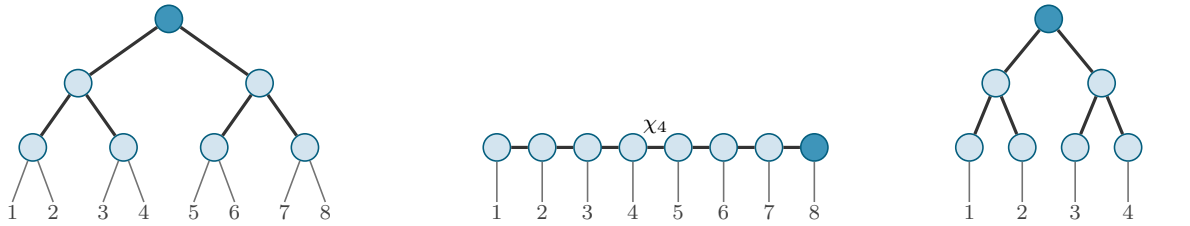
Why this model. It is the simplest genuine TTN. (i) With at least three sites and an internal bond, W is a genuine tensor rather than a matrix; a deep linear network is the special case of a chain with physical legs only at its two ends. (ii) Whitened inputs reduce the population loss to a Frobenius distance (Section 2.3), the exact analogue of the setting in which Saxe et al. (2014) solved deep linear networks, so that theory provides baselines. (iii) It is the regression version of the tensor-network models used in machine learning: the matrix product state (MPS) models of Stoudenmire and Schwab (2016) have the same form $\langle W, \bigotimes_i \phi_i(x) \rangle$, with non-Gaussian local features. (iv) With whitened inputs it is gradient flow on a fully observed hierarchical tensor factorization, whose implicit regularization was studied by Razin et al. (2022).

2.2 Tree topologies

Table 1 lists the built-in topologies and Figure 1 draws them. A binary tree and an MPS with the same N are the two extremes of tree shape at nearly equal K ($N - 1$ versus N) and are the default pair for saddle-to-saddle studies. The hierarchical Tucker (HT) format adds one leaf core per site; when the leaf bond dimension equals d this adds no expressivity, but it changes K and the number of cores that must change to learn a component. Any parent array with a site assignment defines a

Table 1: Built-in tree topologies for N sites (K cores). Depth is the number of edges on the longest root-to-leaf path. Binary-tree values are for $N = 2^m$; for other N the sites are split recursively into halves of sizes $\lceil N/2 \rceil$ and $\lfloor N/2 \rfloor$ until groups of at most two sites remain, each of which becomes a bottom core.

name	structure	K	depth
binary	balanced binary tree; each bottom core carries two adjacent sites, internal cores carry none	$N-1$	$\log_2 N - 1$
MPS	chain (caterpillar), the most unbalanced tree: core k carries site k and its parent is core $k+1$, so the root is core N	N	$N-1$
HT	hierarchical Tucker: balanced binary tree over N leaf cores, one site per leaf core; internal cores carry none	$2N-1$	$\lceil \log_2 N \rceil$
custom	any parent array and site assignment	–	–



(a) binary, $N = 8$, $K = 7$ **(b)** MPS, $N = 8$, $K = 8$ **(c)** HT, $N = 4$, $K = 7$

Figure 1: The built-in tree topologies (Table 1). Circles are cores G_v; the shaded core is the root, whose parent leg (dimension one, the scalar output) is not drawn. Thick lines are bonds: the edge e_v between core v and its parent has dimension χ_v (one label shown in (b), where cores are numbered $1, \ldots, 8$ from the left). Thin dangling lines are physical legs, labelled by their site i (dimension d_i).

custom tree. Unless stated otherwise every edge has the same nominal bond dimension χ, capped at the largest rank its bipartition allows,

$$\chi_v = \min\Big(\chi,\ \prod_{i \in A_v} d_i,\ \prod_{i \notin A_v} d_i\Big). \tag{3}$$

2.3 Teacher, data and loss

Teachers. The target is realizable, $f^*(x) = \langle W^*, \Phi(x) \rangle$. A *CP teacher* has

$$W^* = \sum_{r=1}^{R} s_r U_r, \qquad U_r = u_r^{(1)} \otimes \cdots \otimes u_r^{(N)}, \qquad \|u_r^{(i)}\| = 1, \qquad s_1 \geq \cdots \geq s_R > 0. \tag{4}$$

Its inner products with a TTN do not depend on the TTN's tree (see (9)), and the following special cases have exactly known bond spectra.

- *Odeco* (orthogonally decomposable): $\langle u_r^{(i)}, u_{r'}^{(i)} \rangle = \delta_{rr'}$ at every site, which requires $R \leq \min_i d_i$. By default the factors at each site are columns of an independent Haar-random orthogonal matrix.
- *Pattern teachers*: each site has an orthonormal basis $(\beta_1^{(i)}, \ldots, \beta_{d_i}^{(i)})$ and component r uses $u_r^{(i)} = \beta_{\pi_r(i)}^{(i)}$ for a pattern $\pi_r \colon \{1, \ldots, N\} \to \{1, \ldots, d\}$. Odeco teachers are the case where the

patterns differ at every site; with $\pi_1 = (1,1,1,1)$ and $\pi_2 = (2,2,1,1)$ the second component differs from the first only on the sites $D = \{1,2\}$ (a *local* component).

- *Random CP*: $u_r^{(i)} = g/\|g\|$ with independent $g \sim \mathcal{N}(0, I)$, so the components are generically non-orthogonal.

A *TTN teacher* gives W^* as a TTN on the student's tree or on another tree; a random TTN teacher is a fan-in initialization (Section 2.4) with $\sigma = 1$, rescaled so that $\|W^*\|_F = 1$. The overlap of the student with component r of a CP teacher is

$$a_r = \langle W, U_r \rangle_\Sigma / s_r; \tag{5}$$

for an odeco teacher with whitened inputs, $a_r = 0$ at $W = 0$ and $a_r = 1$ at $W = W^*$.

Proposition 2.1 (Teacher bond spectra)**.** proven *Let e be an edge whose two sides both carry sites (true for every edge of the trees of Table 1). (a) For an odeco teacher the bond spectrum of W^* on e is $(s_1, \dots, s_R)$, for every tree. (b) For a two-component pattern teacher whose components differ exactly on the site set D, the bond rank of W^* on e is 2 if e lies in the smallest subtree of $\mathcal{T}$ containing the nodes that carry the sites of D, and 1 otherwise.*

Proof. Let A be the set of sites on one side of e and write $x_r = \bigotimes_{i \in A} u_r^{(i)}$, $y_r = \bigotimes_{i \notin A} u_r^{(i)}$, so that $(W^*)^{\langle e \rangle} = \sum_r s_r x_r y_r^\top$. (a) Both $(x_r)_r$ and $(y_r)_r$ are orthonormal families, so this is a singular value decomposition. (b) If $D \subseteq A$ or $D \cap A = \emptyset$, then $y_1 = y_2$ or $x_1 = x_2$ and the rank is one. Otherwise the two components differ (by orthogonal basis vectors) on a site of $D \cap A$ and on a site of $D \setminus A$, so $x_1 \perp x_2$, $y_1 \perp y_2$ and the rank is two. Finally, e separates two sites of D exactly when it lies on the tree path between the nodes carrying them, that is, when it belongs to the smallest subtree containing those nodes. □

The number of cores in that subtree is the quantity k of the locality hypothesis tested in Section 4.

Population loss. Training minimizes the population square loss

$$L(\theta) = \tfrac{1}{2}\,\mathbb{E}_x\big[(f_\theta(x) - f^*(x))^2\big] = \tfrac{1}{2}\|W(\theta) - W^*\|_\Sigma^2, \qquad \|A\|_\Sigma^2 = \langle A, \Sigma A \rangle. \tag{6}$$

The second equality holds because, by independence across sites, $\mathbb{E}[\Phi \otimes \Phi] = \bigotimes_i \mathbb{E}[\phi_i \phi_i^\top] = \Sigma$ (as an operator on $\mathbb{R}^{d_1 \times \cdots \times d_N}$). With whitened inputs L is half the squared Frobenius distance. Only second moments enter: every local feature map with $\mathbb{E}[\phi_i \phi_i^\top] = I$ gives the same loss and the same dynamics (for example Hermite features of a scalar input, or $(1, \xi_i)$ with $\xi_i \sim \mathcal{N}(0,1)$), and a non-orthonormal feature map is equivalent to $\Sigma_i = \mathbb{E}[\phi_i \phi_i^\top] \neq I$.

Closed form. Expanding (6),

$$L(\theta) = \tfrac{1}{2}\langle W, W \rangle_\Sigma - \langle W, W^* \rangle_\Sigma + \tfrac{1}{2}\langle W^*, W^* \rangle_\Sigma, \tag{7}$$

and every term is an exact contraction, with no sampling noise:

- For two TTNs W, W' on the same tree (bond dimensions may differ), environments $E_v \in \mathbb{R}^{\chi_v \times \chi'_v}$ are computed from the leaves to the root,

$$(E_v)_{aa'} = \sum G_v[(j_i), (b_c), a]\, G'_v[(j'_i), (b'_c), a'] \prod_{i \in S_v} (\Sigma_i)_{j_i j'_i} \prod_{c \in \mathrm{ch}(v)} (E_c)_{b_c b'_c}, \tag{8}$$

  and $\langle W, W' \rangle_\Sigma = E_{\mathrm{root}}$. The cost is linear in K at fixed d and χ.

- For a CP teacher, $\langle W, U_r\rangle_\Sigma$ is the student evaluated at the site vectors $\Sigma_i u_r^{(i)}$, so for any tree

$$\langle W, W^*\rangle_\Sigma = \sum_r s_r\, f_\theta(\Sigma_1 u_r^{(1)}, \dots, \Sigma_N u_r^{(N)}), \qquad \langle W^*, W^*\rangle_\Sigma = \sum_{r,r'} s_r s_{r'} \prod_i u_r^{(i)\top} \Sigma_i u_{r'}^{(i)}, \quad (9)$$

  where $f_\theta(z_1, \dots, z_N)$ denotes (2) with $\phi_i(x)$ replaced by z_i.

- A TTN teacher on another tree is glued to the student along the physical legs; the resulting network has loops and is contracted exactly along an optimized contraction path.

For an odeco teacher with whitened inputs, (7) reads $L = \frac{1}{2}\|W\|_F^2 - \sum_r s_r^2 a_r + \frac{1}{2}\sum_r s_r^2$. A *partial fit* $W_S = \sum_{r\in S} s_r U_r$ that has learned exactly the components in S has loss $\frac{1}{2}\sum_{r\notin S} s_r^2$; in particular, learning the components in decreasing order of s_r passes through the levels

$$L_k = \tfrac{1}{2}\sum_{r>k} s_r^2, \qquad k = 0, \dots, R. \quad (10)$$

These are the candidate plateau levels examined in Section 4.

Anisotropic and correlated inputs. Anisotropic product covariances are handled exactly by (6)–(9); the default family has power-law site spectra, $\Sigma_i = c_\beta\, \mathrm{diag}(1^{-\beta}, 2^{-\beta}, \dots, d^{-\beta})$ with c_β fixed by $\mathrm{tr}\,\Sigma_i = d$, aligned with the teacher's factors or rotated away from them. Cross-site correlations break the product structure of Σ; they are studied with sampled data only (Section 6).

Finite samples. For sample-size questions, a training set $(x_j, y_j)_{j=1}^n$ has $x_j \sim \mathcal{N}(0, \Sigma)$ i.i.d. and $y_j = f^*(x_j) + \varepsilon_j$ with optional label noise $\varepsilon_j \sim \mathcal{N}(0, \varsigma^2)$. The empirical loss is $\hat{L}_n(\theta) = \frac{1}{2n}\sum_j (f_\theta(x_j) - y_j)^2$, and the test error is the population loss (6).

2.4 Initialization and its variants

Fan-in initialization. The default initialization is

$$G_v = \frac{\sigma_v}{\sqrt{F_v}}\, Z_v, \qquad Z_v \text{ with i.i.d.} \mathcal{N}(0,1) \text{ entries}, \quad (11)$$

with a single per-core scale $\sigma_v = \sigma$ unless stated otherwise. Each core is then approximately σ_v times an isometry (when $F_v \gg \chi_v$), so that with $\sigma_v = 1$ the messages (2) keep their size on the way up the tree.

Proposition 2.2 (Initial scales). proven *Under* (11), $\mathbb{E}\|W(\theta_0)\|_F^2 = \prod_v \sigma_v^2$ *exactly,* $\mathbb{E}\|G_v\|_F^2 = \sigma_v^2 \chi_v$, *and hence* $\mathbb{E}\|\theta_0\|^2 = \sum_v \sigma_v^2 \chi_v$.

Proof. Let T_v be the contraction of the subtree below v with its parent leg open, a matrix with χ_v columns. Conditionally on the children's subtrees, the independence and variance σ_v^2/F_v of the entries of G_v give $\mathbb{E}[T_v^\top T_v] = \frac{\sigma_v^2}{F_v}\prod_{i\in S_v} d_i \prod_{c\in \mathrm{ch}(v)} \mathrm{tr}(T_c^\top T_c)\, I_{\chi_v}$. The children's subtrees are independent, so induction from the leaves gives $\mathbb{E}[T_v^\top T_v] = (\prod_u \sigma_u^2) I_{\chi_v}$, the product running over the subtree of v, because $F_v = \prod_{i\in S_v} d_i \prod_c \chi_c$. At the root $\chi_{\mathrm{root}} = 1$ and T_{root} is W flattened. The second claim is $\mathbb{E}\|G_v\|_F^2 = F_v \chi_v\, \sigma_v^2/F_v$. □

We therefore report both the per-core scale σ and the function-space scale $\alpha = \sigma^K$, the root-mean-square norm of $W(\theta_0)$. Small σ makes $W(\theta_0)$ exponentially small in K, while $\|\theta_0\|^2 \approx \sigma^2 \sum_v \chi_v$ grows with the bond dimensions.

Variants.

(a) *Imbalance*: per-core scales $\sigma_v = \sigma c_v$ with a fixed profile (c_v), for example root versus the rest, or by depth, compared at fixed function-space scale $\prod_v \sigma_v$.

(b) *Centred model*: the target is $W^* + W(\theta_0)$, that is, $f_\theta - f_{\theta_0}$ is trained on f^* (Chizat et al., 2019). The initial function is removed exactly at any σ without changing the parameterization, so σ alone moves the model from small to large initialization. Overlaps are measured on $W - W(\theta_0)$. This is the main construction of the regime map (Section 3).

(c) *Two-copy model* (Woodworth et al., 2020): $W(\theta^+) - W(\theta^-)$ with $\theta^+(0) = \theta^-(0) = \theta_0$, which vanishes at initialization for every σ. It is represented as one TTN on the same tree with doubled bond dimensions $2\chi_v$ and block-diagonal cores: core v holds G_v^+ on the block where all its bond indices are in the "+" half and G_v^- on the "−" block, except that the root holds G_{root}^+ and $-G_{\text{root}}^-$. It is used as a cross-check of the centred model; see Remark 2.3.

(d) *Aligned (decoupled) initialization* for CP teachers (Proposition 2.4): cores diagonal in the teacher's basis, giving exactly solvable Saxe-type dynamics.

(e) *Per-core learning-rate multipliers* η_v (Section 2.5). A fixed forward multiplier c_v on core v (writing $G_v = c_v H_v$ and training H_v) is equivalent to the initial scale $c_v\sigma_v$ together with the multiplier $c_v^2\eta_v$. This is how NTK-type and μP-type parameterizations for large bond dimension (Yang and Hu, 2021) are expressed (Section 5).

Remark 2.3 (The two-copy model needs a gradient mask). proven Gradient flow on the doubled TTN does not preserve its block-diagonal structure. For a root with children c_1, c_2, the gradient of L with respect to the root entry pairing index b_1 of the + half of bond e_{c_1} with index b_2 of the − half of bond e_{c_2} is $\langle W - W^*,\, T_{c_1}^+[:, b_1] \otimes T_{c_2}^-[:, b_2]\rangle_\Sigma$ (up to the ordering of sites), which at initialization equals $-\langle W^*,\, T_{c_1}^+[:, b_1] \otimes T_{c_2}^-[:, b_2]\rangle_\Sigma$ and is generically nonzero; mixed blocks of non-root cores also receive gradient. The two-copy dynamics is therefore the masked flow $\dot\theta = -m \odot \nabla L(\theta)$, with m the 0/1 indicator of the diagonal blocks, which keeps the blocks and equals gradient flow of $\frac{1}{2}\|W(\theta^+) - W(\theta^-) - W^*\|_\Sigma^2$ in (θ^+, θ^-).

Proposition 2.4 (Aligned initialization). proven known (deep linear networks: Saxe et al., 2014) *Let W^* be odeco, let inputs be whitened, let $\chi_v \geq R$ for every edge, and let every subtree and every subtree's complement carry at least one site (true for the trees of Table 1). Initialize*

$$G_v = \sum_{r=1}^{R} q_{v,r} \Big(\bigotimes_{i \in S_v} u_r^{(i)} \Big) \otimes \Big(\bigotimes_{c \in \mathrm{ch}(v)} \varepsilon_r \Big) \otimes \varepsilon_{r(v)}, \tag{12}$$

with ε_r the r-th standard basis vector of the bond in question, $r(v) = r$ for $v \neq \text{root}$ and $r(\text{root}) = 1$. Under gradient flow with multipliers η_v, the cores keep the form (12), $W = \sum_r \Pi_r U_r$ with $\Pi_r = \prod_v q_{v,r}$, $a_r = \Pi_r / s_r$, $L = \frac{1}{2}\sum_r (s_r - \Pi_r)^2$, and

$$\dot q_{v,r} = \eta_v \, (s_r - \Pi_r) \prod_{u \neq v} q_{u,r}. \tag{13}$$

With $\eta_v = 1$ the differences $q_{v,r}^2 - q_{u,r}^2$ are conserved; a balanced start $q_{v,r}(0) = q_r(0)$ stays balanced, and $\dot q_r = q_r^{K-1}(s_r - q_r^K)$.

Proof. On the family (12), contracting the basis vectors on the bonds gives $W = \sum_r \Pi_r U_r$, and the U_r are orthonormal. The derivative of W with respect to an entry of G_v is the tensor product of

basis vectors on the physical legs of v, the columns $T_c[:, b_c]$ of the child subtrees and the column of the environment above v indexed by the parent index a. On the family these columns are multiples of $\bigotimes_{i\in A_c} u_{b_c}^{(i)}$ (and of the analogous product over the sites outside A_v), so the inner product with $W - W^* = \sum_r (\Pi_r - s_r) U_r$ vanishes unless all bond indices equal the same r (with $a = 1$ at the root). The gradient therefore lies in the span of the family, and its component along $q_{v,r}$ is $(\Pi_r - s_r) \prod_{u\neq v} q_{u,r}$. Finally, with $\eta_v = 1$, $\frac{\mathrm{d}}{\mathrm{d}t} q_{v,r}^2 = 2(s_r - \Pi_r)\Pi_r$ is the same for every v. □

For $K \geq 3$ a balanced mode started at $q_r(0) \ll s_r^{1/K}$ reaches $a_r = \frac{1}{2}$ at

$$t_r = \frac{q_r(0)^{-(K-2)}}{(K-2)\, s_r} + O(s_r^{-(2-2/K)}), \tag{14}$$

and for $K = 2$ at $t_r = \log(s_r/q_r(0)^2)/(2s_r) + O(q_r(0)^2/s_r^2)$ (direct integration of the balanced ODE) proven. With $q_r(0) \propto \sigma$ the plateau before the first escape grows like $\sigma^{-(K-2)}$; this time scale is why the integration below must be adaptive and in double precision.

2.5 Training dynamics

Gradient flow is the reference dynamics,

$$\dot\theta = -D\, \nabla L(\theta), \tag{15}$$

where D is diagonal with entry $\eta_v m_j$ for every parameter j of core v: the η_v are the per-core learning-rate multipliers (default 1) and $m \in \{0,1\}^{\dim\theta}$ is an optional mask (Remark 2.3; default all ones). Here t always denotes gradient-flow time. It is integrated with the adaptive Dormand–Prince 5(4) pair (Dormand and Prince, 1980) (first-same-as-last, local extrapolation) in double precision, with relative and absolute tolerances 10^{-8} and 10^{-12} by default. A step of size h is accepted when the root-mean-square over components of the embedded error estimate, each divided by $10^{-12} + 10^{-8}\max(|\theta_j|, |\theta_j^{\mathrm{new}}|)$, is at most one; the next step is $h \min\{5, \max\{0.2, 0.9\,\mathrm{err}^{-1/5}\}\}$, with no increase right after a rejection, and the first step follows Hairer et al. (1993, Sec. II.4). Step sizes thus follow the local time scale $\|\theta\|/\|\dot\theta\|$, so that long plateaus are cheap. Diagnostics are recorded at $t = 0$ and at 200 log-spaced times from $t_{\min} = 10^{-3}\min(\tau_0, t_{\max})$ to $t_{\max}$, where $\tau_0 = \|\theta_0\|/\|\dot\theta(0)\|$; steps are shortened to land on these times exactly. A run stops at $t_{\max}$, when L falls below a prescribed floor, after a maximum number of steps, or if the step size underflows. Accuracy is checked with the conservation laws (Section 2.6) and against the closed-form aligned solutions (Proposition 2.4).

Gradient descent with learning rate η, $\theta_{k+1} = \theta_k - \eta D \nabla L(\theta_k)$, is recorded at $t = \eta k$, so it approaches gradient flow at equal t as $\eta \to 0$. A run has diverged when L becomes non-finite or exceeds $10^6 L(\theta_0)$.

Stochastic gradient descent uses the minibatch loss $\hat L_B(\theta) = \frac{1}{2B}\sum_{b=1}^{B}(f_\theta(x_b) - y_b)^2$ in the same update, again recorded at $t = \eta k$. Online SGD draws fresh inputs $x_b \sim \mathcal{N}(0, \Sigma)$ with $y_b = f^*(x_b)$ at every step; with a fixed training set, batches are taken from a new random permutation in every epoch. Optional label noise is drawn afresh at every step.

Precision. All computations are in double precision: with $\sigma = 10^{-2}$ and $K = 7$, $\|W(\theta_0)\|_F \approx 10^{-14}$ and the gradients are of order 10^{-12}, which single precision cannot resolve over long plateaus.

2.6 Symmetries and conservation laws

For a bond $e = e_v$ joining core v to its parent $p = p(v)$, let $G_{u,(e)}$ denote core $u \in \{v, p\}$ matricized with its e-leg as rows (a $\chi_v \times F_v$ matrix for $u = v$). For every invertible $B \in \mathrm{GL}(\chi_v)$ the *gauge transformation*

$$G_{v,(e)} \mapsto B\, G_{v,(e)}, \qquad G_{p,(e)} \mapsto B^{-\top} G_{p,(e)} \tag{16}$$

leaves W, and hence any loss that depends on θ only through W, unchanged.

Proposition 2.5 (Conservation laws). proven known (deep linear networks: Arora et al., 2018; Du et al., 2018) *Under gradient flow $\dot{G}_u = -\eta_u \nabla_{G_u} L$ of any loss that depends on θ only through W (population or empirical), the $\chi_v \times \chi_v$ matrix*

$$Q_e = \frac{1}{\eta_v} G_{v,(e)} G_{v,(e)}^\top - \frac{1}{\eta_p} G_{p,(e)} G_{p,(e)}^\top \tag{17}$$

is constant in time, for every bond $e = e_v$ with $p = p(v)$.

Proof. Write $X = G_{v,(e)}$, $Y = G_{p,(e)}$ and take $B = I + \epsilon M$ in (16), so that $X \mapsto X + \epsilon M X$ and $Y \mapsto Y - \epsilon M^\top Y + O(\epsilon^2)$. Invariance of L at first order in ϵ gives $\langle \nabla_X L\, X^\top - Y\,(\nabla_Y L)^\top,\, M \rangle = 0$ for every M, so $C := \nabla_X L\, X^\top = Y (\nabla_Y L)^\top$. Along the flow, $\frac{\mathrm{d}}{\mathrm{d}t}(XX^\top)/\eta_v = -(C + C^\top) = \frac{\mathrm{d}}{\mathrm{d}t}(YY^\top)/\eta_p$. □

This is the tree version of the balancedness of deep linear networks and an instance of the general correspondence between continuous symmetries and conserved quantities of gradient flow (Kunin et al., 2021). Consequences:

- *Norm balance*: $\mathrm{tr}\, Q_e = \|G_v\|_F^2/\eta_v - \|G_p\|_F^2/\eta_p$. With equal multipliers, the differences $\|G_u\|_F^2 - \|G_{u'}\|_F^2$ between any two cores are conserved (the tree is connected), so all cores grow together.
- *Balancedness at small initialization*: under (11), $\mathbb{E}\, Q_e(0) = (\sigma_v^2/\eta_v - \sigma_p^2 \chi_p/(\eta_p \chi_v))I$, so with small σ the cores remain balanced up to $O(\sigma^2)$ while they grow to $O(1)$. Imbalanced initializations or multipliers set Q_e to other values, a second knob for the regime map (Kunin et al., 2024).
- The masked two-copy flow conserves the Q_e of each copy (its gauge group is the block-diagonal subgroup). Gradient descent and SGD conserve Q_e only up to $O(\eta^2)$ per step.
- *Integrator check*: the conservation drift

$$\mathrm{drift}(t) = \frac{\max_e \|Q_e(t) - Q_e(0)\|_{\max}}{\max_e \|Q_e(0)\|_{\max} + \max_u \|G_u(t)\|_F^2/\eta_u}, \tag{18}$$

  with $\|\cdot\|_{\max}$ the largest absolute entry, measures the integration error relative to the size of the terms that cancel in Q_e. It must stay below 10^{-6} under gradient flow.

2.7 Regime diagnostics

Regimes are assigned from measured quantities, never by eye. The thresholds below are conventions; every phase diagram reports how its boundaries move when they change.

Raw quantities (Table 2) are recorded on the logarithmic time grid of Section 2.5, with extra points where needed. The tangent kernel uses a fixed probe set of $P = 128$ inputs $x_1, \ldots, x_P \sim \mathcal{N}(0, \Sigma)$ and the Jacobian $J_t = \partial(f_\theta(x_1), \ldots, f_\theta(x_P))/\partial\theta$ at $\theta(t)$. The gauge directions at θ are the tangent vectors of the orbits (16), $(\delta G_{v,(e)}, \delta G_{p,(e)}) = (M G_{v,(e)}, -M^\top G_{p,(e)})$ for all bonds e and $M \in \mathbb{R}^{\chi_v \times \chi_v}$; $H_\perp$ is the Hessian restricted to their orthogonal complement.

Table 2: Raw diagnostics recorded along every run.

symbol	definition
$L(t)$, $\ell(t)$	exact population loss (6); normalized loss $\ell(t) = L(t)/L(0)$
$s_k^e(t)$	bond spectrum of $W(t)$ across edge e; function-space version when $\Sigma \neq I$
$s_k^{e,*}$	bond spectrum of W^*; equal to the weights s_k on every edge for an odeco teacher (Proposition 2.1)
$a_r(t)$	overlap (5) with teacher component r (CP teachers); of $W - W(\theta_0)$ in the centred model
$\delta_\theta(t)$	$\|\theta(t) - \theta(0)\|/\|\theta(0)\|$, also per core
Θ_t	tangent kernel $J_t J_t^\top$ on the probe set; for small N also the tensor-space operator $\sum_v (\partial W/\partial G_v)(\partial W/\partial G_v)^\top$
$\delta_\Theta(t)$	relative kernel change $\|\Theta_t - \Theta_0\|_F/\|\Theta_0\|_F$
$\mathcal{A}(t)$	kernel-target alignment $y^\top \Theta_t y/(\|\Theta_t\|_F \|y\|^2)$, with $y_p = f^*(x_p)$ on the probe set
drift(t)	conservation drift (18), an integrator check
$H(t)$, $H_\perp(t)$	Hessian of L at $\theta(t)$, and its restriction to the complement of the gauge directions

Learning events. For a CP teacher, event r occurs at the first time t_r at which $a_r(t)$ crosses $\frac{1}{2}$, and its width is $w_r = t(a_r = 0.9) - t(a_r = 0.1)$. For a general teacher, event (e, k) occurs when $s_k^e(t)$ first crosses half of its final value, with widths defined in the same way from the crossings of 10% and 90% of the final value; since one component typically raises several bond spectra at once, events closer than their widths are merged. The *effective bond rank* is $\mathrm{rk}_e(t) = \#\{k : s_k^e(t) > 0.1\, s_{\min}^{e,*}\}$, where $s_{\min}^{e,*}$ is the smallest nonzero teacher Schmidt value on e; the threshold-free alternative is the entropy rank $\mathrm{rk}_e^H = \exp(-\sum_k p_k \log p_k)$ with $p_k = (s_k^e)^2/\sum_j (s_j^e)^2$.

Plateaus and the saddle-to-saddle signature. Number the events in time order, $t_1 < t_2 < \cdots$, with widths w_i, and define the *separation ratios*

$$\rho_i = \frac{t_{i+1} - t_i}{\max(w_i, w_{i+1})}. \tag{19}$$

Between two events with $\rho_i > 3$ the loss sits on a plateau at the level of a partial fit. The plateau midpoint t_i^{mid} is the time of minimal speed $\|\dot\theta(t)\|$ in $[t_i, t_{i+1}]$, and the Hessian is evaluated there. A run shows the *saddle-to-saddle (S2S) signature* when

(i) there are at least two events,

(ii) every separation ratio satisfies $\rho_i > 3$, and

(iii) across a sweep, plateau durations grow without bound as the initialization scale tends to zero (like $\log(1/\sigma)$ or a power of $1/\sigma$).

Plateaus are judged in linear time between events, not only on logarithmic time axes: an escape that needs two cores takes a time of order $\log(1/\sigma)$, which can be invisible on a log-time axis after a long first plateau.

Regime assignment. Each run, trained up to time T, receives one label by the rules of Table 3.

Remark 2.6 (Reading of the rules). "The loss converges" means $L(T) < 10^{-3} L(0)$, the complement of "stalled". Labels are checked in the order diverged, stalled, then the three converged regimes. Condition (iii) is a property of a sweep: a single run is labelled S2S on the basis of (i)–(ii), and

Table 3: Regime assignment for one run trained up to time T (the S2S signature is defined in Section 2.7). Threshold sensitivity: the kernel threshold 0.1 is varied over $\{0.05, 0.1, 0.2\}$ and the separation threshold 3 over $\{2, 3, 5\}$.

regime	rule
lazy	the loss converges and $\delta_\Theta(T) < 0.1$ (so δ_θ is small as well)
saddle-to-saddle	$\delta_\Theta(T) \geq 0.1$ and the S2S signature (i)–(iii)
rich, non-incremental	$\delta_\Theta(T) \geq 0.1$ and the loss converges, but some $\rho_i \leq 3$ (overlapping transitions)
stalled	$L(T) \geq 10^{-3} L(0)$ within the time budget (for example a long initial plateau)
diverged	GD and SGD only: L non-finite or above $10^6 L(0)$

a region of a phase diagram is reported as S2S only if (iii) holds there. A converged run with $\delta_\Theta(T) \geq 0.1$ and a single (merged) event is counted as rich, non-incremental.

Saddle type at a plateau. At a plateau midpoint, a *strict saddle* has $\lambda_{\min}(H_\perp) < -\epsilon$ with $\epsilon = 10^{-3} \max_j |\lambda_j(H_\perp)|$; a *higher-order (non-strict) saddle* has no eigenvalue below $-\epsilon$ although the loss later decreases. We report the number of negative eigenvalues, $\lambda_{\min}$ (also unnormalized) and $\lambda_{\min}/\lambda_{\max}$.

Reporting rules. Every scaling claim uses at least five initialization scales spanning at least one decade and at least five seeds per point, reports mean ± standard error and a fitted exponent with its standard error, and overlays the theoretical prediction. Every regime boundary comes with the threshold sensitivity check of Table 3. Every number states the dynamics that produced it (gradient flow with its tolerances, GD with its step size, or SGD with its step size and batch size).

2.8 What the results may depend on

- *Population versus finite samples.* Plateaus and saddle sequences are properties of the population gradient flow; statements about implicit bias and generalization need finite samples and are tested separately.
- *Multilinear features.* There is no pointwise nonlinearity; orthonormal local feature maps all give the same population dynamics, and non-orthonormal ones are equivalent to anisotropic Σ_i.
- *Product input measure.* It makes the population loss a tree contraction; cross-site correlations break this and are studied with sampled data.
- *Placement of physical legs.* Binary trees (two sites per bottom core) and HT (one leaf core per site) differ in K and in how many cores must change to learn a component, which is expected to change plateau-time exponents.
- *Scalar output.* Multi-output regression or classification would add an output leg to the root core.
- *Fan-in initialization.* It fixes the relative scales of the cores. Exponents should not depend on it, but prefactors and the order of learning might, which is why imbalanced and aligned initializations are also studied.

3 Map of regimes

[Skeleton. Phase diagrams of the training regimes over initialization scale, bond dimension, number of cores and core imbalance, with predicted boundaries and threshold sensitivity. Waits for the regime-map results.]

3.1 Protocol

[Centred model as the one-knob family; labels from Table 3; seeds, sweeps, time budgets.]

3.2 Initialization scale and bond dimension

[Phase diagram $\sigma \times \chi$ for binary trees with $N = 4, 8$.]

3.3 Number of cores and function-space scale

[Phase diagram $\sigma \times K$; boundaries in both σ and $\alpha = \sigma^K$.]

3.4 Imbalance between cores

[Does the conserved imbalance Q_e move the TTN between regimes, as in deep linear networks?]

3.5 Large bond dimension

[NTK-type versus μP-type parameterizations as χ grows.]

3.6 Threshold sensitivity and cross-checks

[How the boundaries move with the kernel threshold $\{0.05, 0.1, 0.2\}$ and the separation threshold $\{2, 3, 5\}$; two-copy cross-check.]

3.7 Summary: which parameter moves which boundary

[Table: parameter, boundary moved, direction, fitted versus predicted.]

4 The saddle-to-saddle regime

[Skeleton. Theory and experiments for gradient flow from small initialization: plateau sequence, escape times and their dependence on each parameter, locality of later escapes, learning order, saddle types.]

4.1 Phenomenology

[Representative runs: loss in linear and logarithmic time, overlaps, bond spectra on every edge, effective ranks; S2S signature (Section 2.7).]

4.2 An exactly solvable case: aligned initialization

[Proposition 2.4 in experiments: TTN gradient flow against the exact mode solutions for binary, MPS and HT trees.]

4.3 Escape from the origin

[First escape time t_1 versus σ, χ and K, against the prediction.]

4.4 Later saddles and locality

[Escape times after the first plateau depend on where the new component lives in the tree.]

4.5 Order of learning

[Which component is learned first, how random the order is, and how a wrong order is corrected.]

4.6 Dependence on each parameter

[Summary table: initialization scale, number of cores and tree shape, bond dimension, physical dimension, teacher spectrum, imbalance, learning rate; effect on plateau levels, durations and order; status of each statement.]

5 Other regimes

[Skeleton. The regimes other than saddle-to-saddle, and how discrete steps, gradient noise and finite samples modify the gradient-flow picture.]

5.1 Lazy regime

[Large initialization in the centred and two-copy models: linearized dynamics, kernel at initialization, time rescaling by $\sigma^{2(K-1)}$.]

5.2 Rich, non-incremental regime

[Intermediate scales where features move but transitions overlap; what sets the crossover.]

5.3 Stalled runs

[Runs that do not converge within the budget: long initial plateaus at small σ and large K, and the relation to barren plateaus.]

5.4 Large bond dimension: NTK versus μP

[Per-core scalings for an infinite-bond-dimension limit with and without feature learning.]

5.5 Discrete steps, gradient noise and finite samples

[Gradient descent versus gradient flow, the stability limit and edge of stability; SGD noise and plateau durations; sample complexity of the saddle-to-saddle and lazy regimes.]

6 Effects of input covariance and teacher structure

[Skeleton. Starting from whitened inputs and an odeco teacher, one complication at a time: does it change the saddle-to-saddle picture qualitatively, quantitatively, or not at all?]

6.1 Anisotropic covariance aligned with the teacher

[Prediction: component r behaves as if its weight were $\Lambda_r s_r$, $\Lambda_r = \prod_i \lambda_{r,i}$.]

6.2 Misaligned covariance

[Teacher factors that are not covariance eigenvectors.]

6.3 Cross-site correlations

[One correlated input model (shared latent factor), with sampled data and small steps.]

6.4 Non-orthogonal CP teachers

[Saddle sequence and plateau levels versus the overlap between components.]

6.5 Random TTN teachers

[Are the visited saddles the hierarchical-SVD truncations of the teacher, and in which order do bond ranks grow?]

6.6 Teacher and student on different trees

[Which bonds grow first when the teacher's correlation structure does not match the student's tree; connection to locality (Section 4.4).]

6.7 Summary

[Table: complication, why it was checked, what changed, and whether the baseline theory (possibly after a substitution such as effective weights) predicts it.]

7 Open questions

[Skeleton. Questions left open by this work, each with what would settle it.]

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A Derivations

[Skeleton. Full derivations behind the statements of the main text; the short proofs of Section 2 are not repeated.]

A.1 Gradients and the tangent kernel of a TTN

[Environment formula for $\nabla_{G_v} L$; the tangent kernel at initialization and its scale $\sigma^{2(K-1)}$.]

A.2 Escape from the origin

[Spherical flow and radial growth for the degree-K homogeneous correlation $g(\theta) = \langle W(\theta), W^* \rangle$; alignment time; the first-escape formula of Section 4.3.]

A.3 Expansion around a learned-component saddle

[Second-order expansion of L in the directions that add a new component; the locality rule for k and the saddle type.]

A.4 Correction of a wrong learning order

[The "piggyback" direction and its Hessian eigenvalue $q(s_{\text{learned}} - s_{\text{new}})$ (hypothesis H1b).]

A.5 Gradient-descent stability limit

[Sharpness at the balanced global minimum and the predicted $\eta_{\max}$.]

A.6 Regime boundaries

[Lazy criterion and saddle-to-saddle separation criterion as functions of σ, χ, K and imbalance; large-bond-dimension scalings.]

B Experimental details

[Skeleton. For every experiment: trees, dimensions, teachers, initialization, dynamics and tolerances, sweeps, seeds, time budgets, compute cost.]

B.1 Common settings

[Double precision; Dormand–Prince 5(4) with relative and absolute tolerances 10^{-8} and 10^{-12} unless stated; recording grid; conservation-drift threshold 10^{-6}; event detection; Hessian and kernel computations.]

B.2 Saddle-to-saddle experiments

[Settings of the experiments of Section 4 (s2s-core, locality, order).]

B.3 Regime-map experiments

[Grids, seeds and time budgets of the phase diagrams of Section 3.]

B.4 Covariance and teacher experiments

[Settings of Section 6.]

B.5 Optimizer and sample-size experiments

[Learning rates, batch sizes, training-set sizes and seeds of Section 5.5.]

C Reproducibility

[Skeleton. How every number and figure in this report is regenerated.]

C.1 Software and data

[Repository layout, environment and where raw outputs live.]

C.2 Figures

[Table: figure, experiment folder, script and command, config, raw data needed.]

C.3 Building the report

The report is built by `report/build.sh` (pdflatex, BibTeX, then pdflatex until the cross-references are stable), after `report/collect_figures.sh` has copied the figures listed in `report/figures.txt` from `experiments/*/results/` into `report/figures/`. `report/README.md` describes both scripts.